

In practice, I used a MCP2515 module for SPI-CAN conversion

In practice, I use STM32F411CEU6 with 5 timer, STM32F401 will not have enough timer for 4 encoder+PWM. pin B11 on the schematic would be pin B2 on STM32F411

Wire Color:
Black: Ground

Red:12V

Orange:5V

Yellow:3.3V

White: Clock/PWM

Grey: GPIO Output

Purple/Light Brown: Input

Dark Brown: Output

Blue&Green: H-Bridge 12V Motor input

Cyan&Pink: 5V CAN communication lines

Used WaveShare CAN HAT in practice

